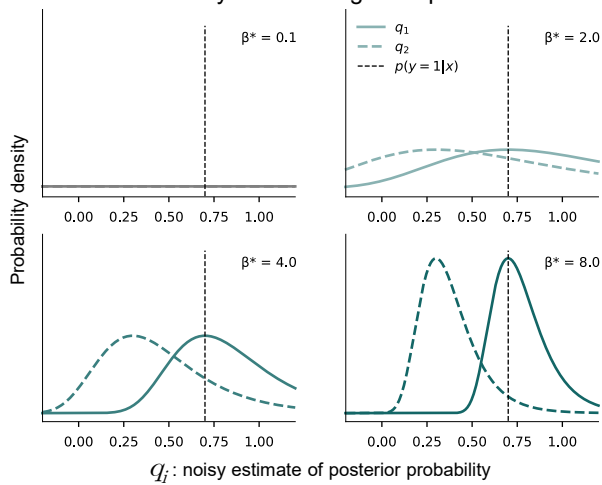
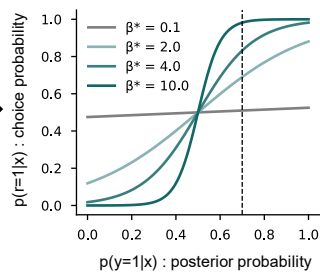
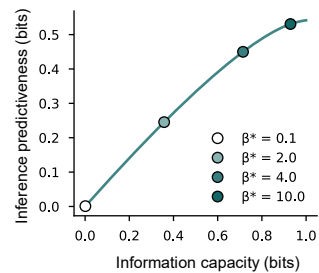


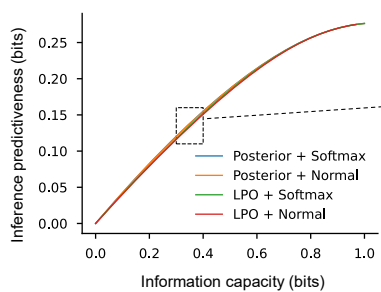
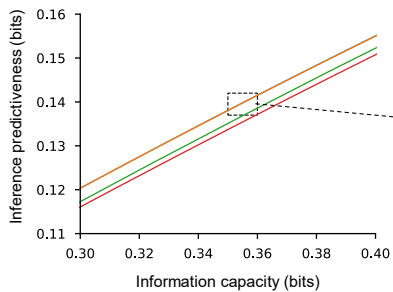
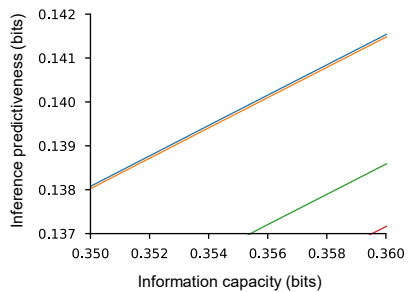
A

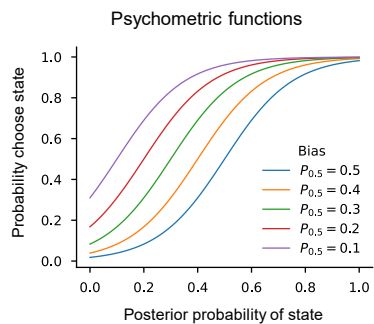
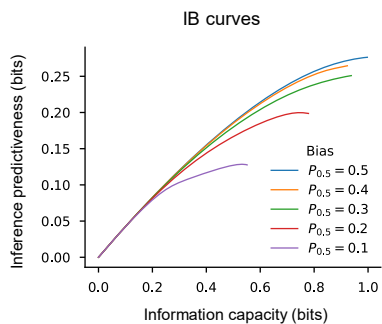
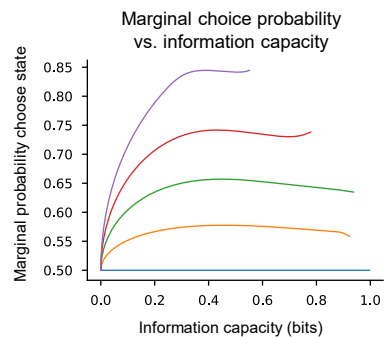
Noise in estimate of posterior probabilities,
Always choose highest option

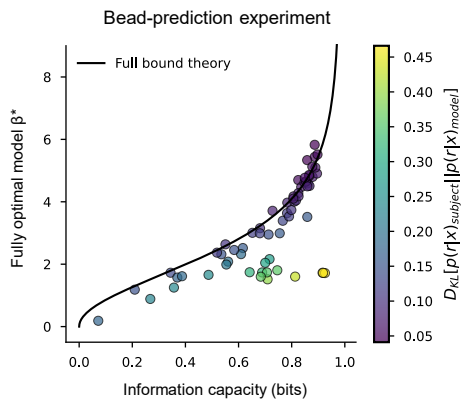
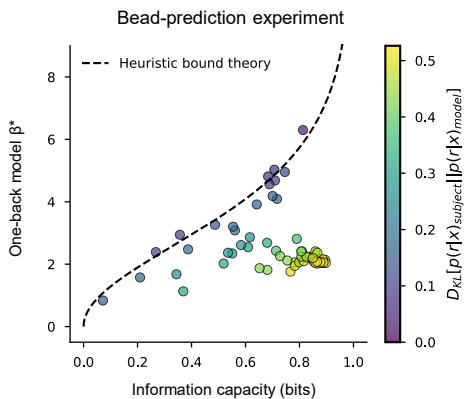
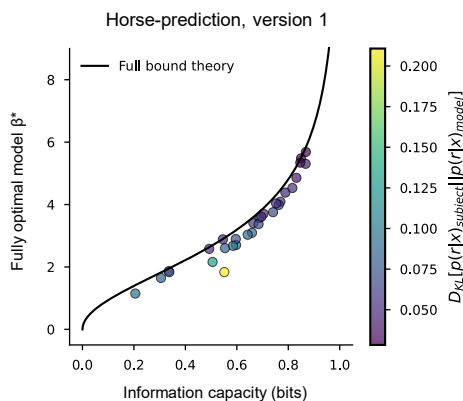
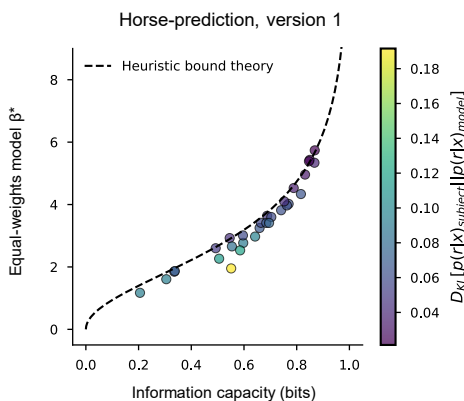
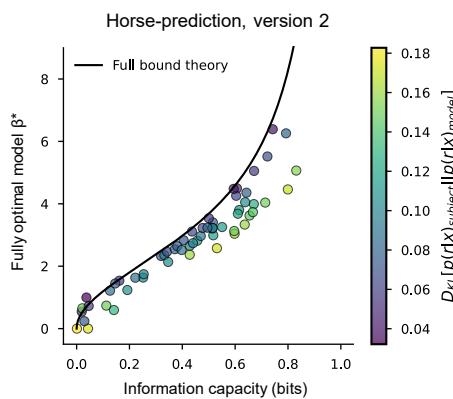
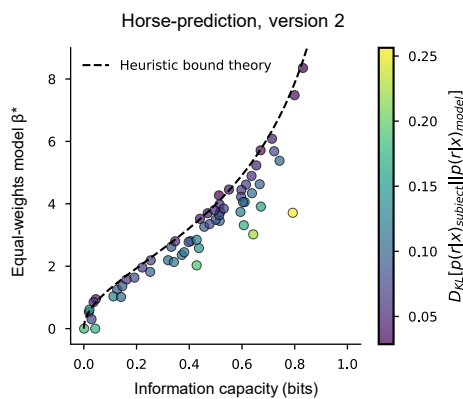
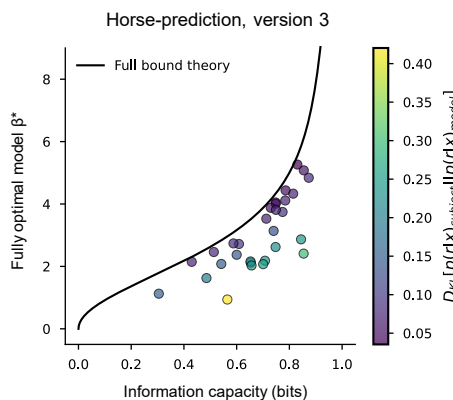
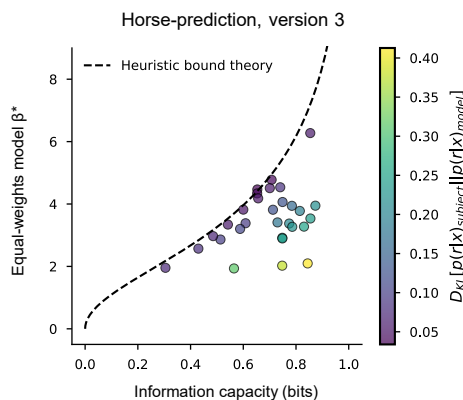
**B**

Evidence-based
choice noise

**C**

A**B****C**

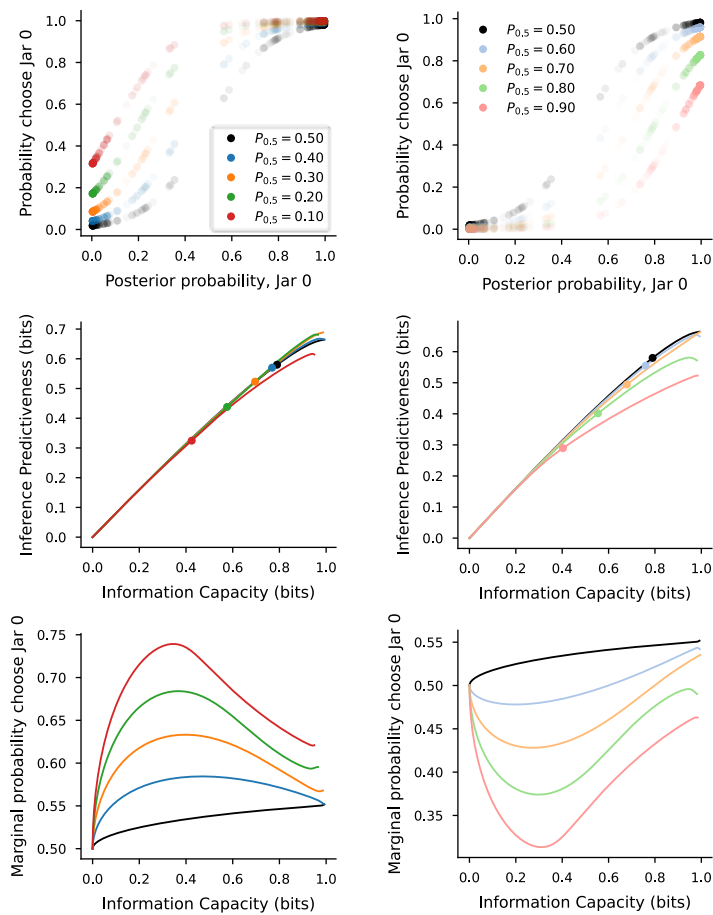
A**B****C**

A**B****C****D****E****F****G****H**

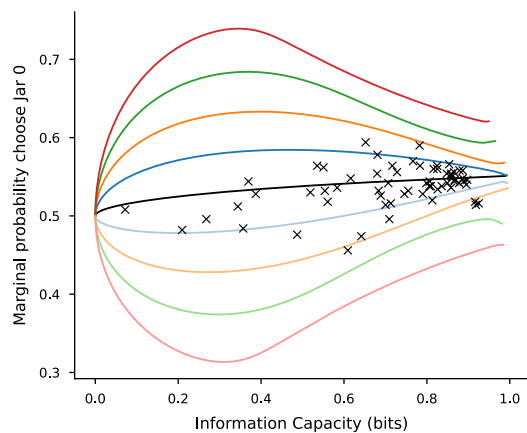
A

Bead prediction task

Simulations with different amounts of marginal choice bias

**B**

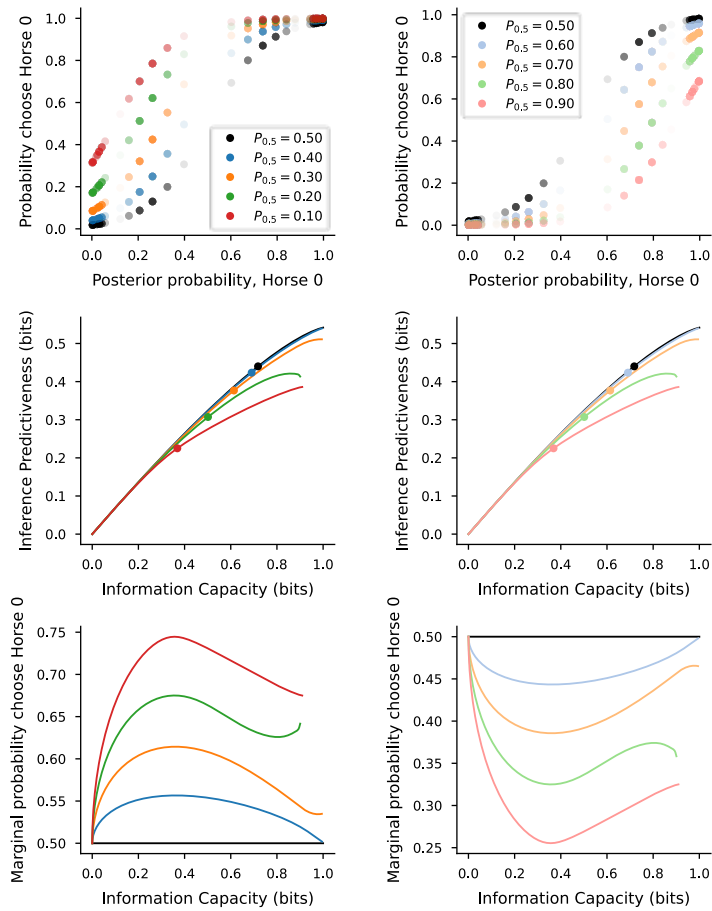
Participant marginal choice probabilities



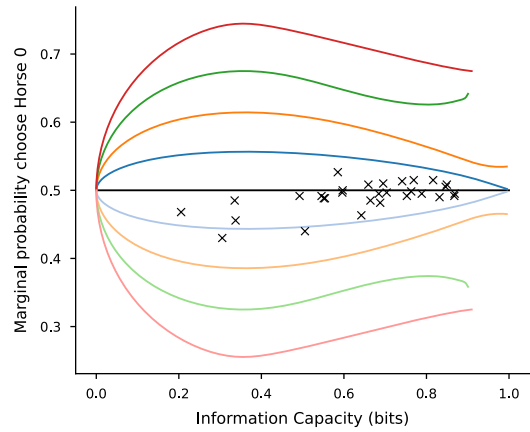
A

Horse prediction, low W-S ratio (1.3:1)

Simulations with different amounts of marginal choice bias

**B**

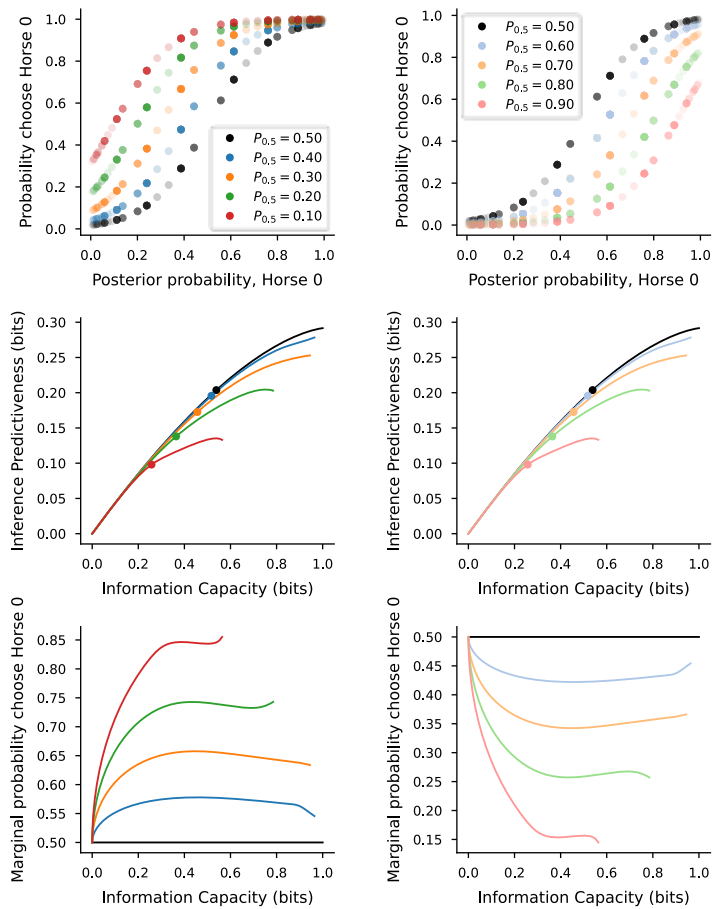
Participant marginal choice probabilities



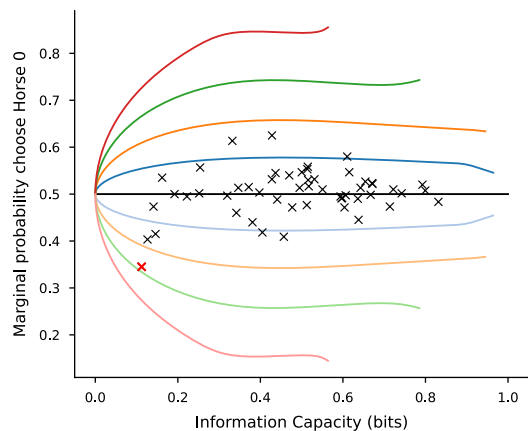
A

Horse prediction, intermediate W-S ratio (2.5:1)

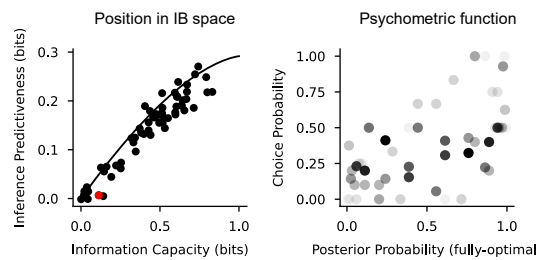
Simulations with different amounts of marginal choice bias

**B**

Participant marginal choice probabilities

**C**

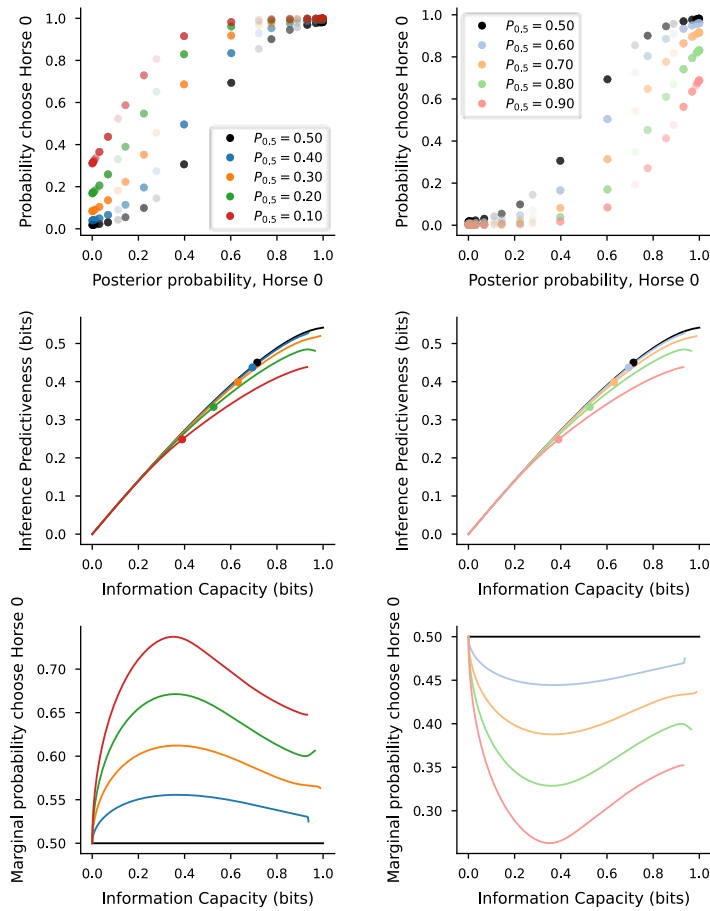
Participant with unusually high bias (red X above)



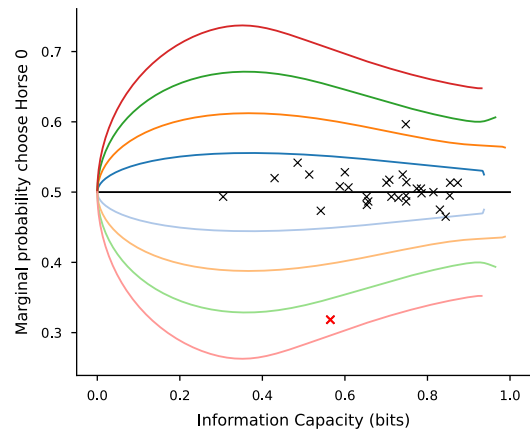
A

Horse prediction, high W-S ratio (6.3:1)

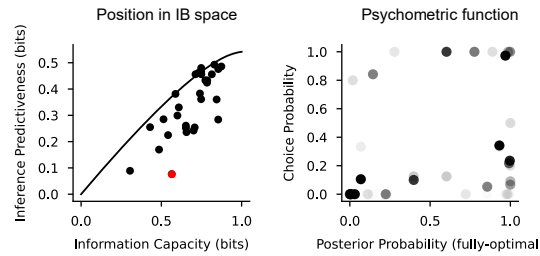
Simulations with different amounts of marginal choice bias

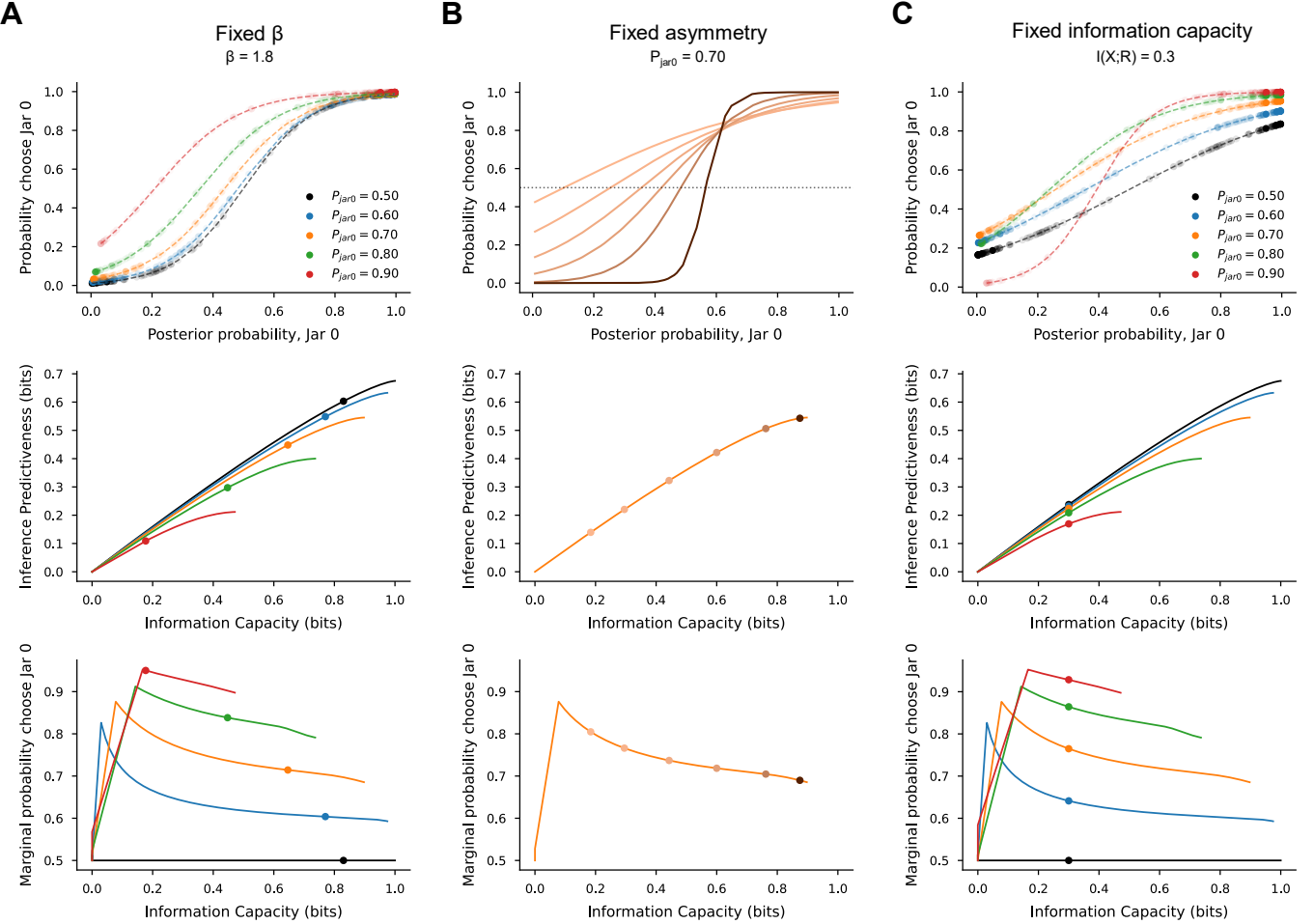
**B**

Participant marginal choice probabilities

**C**

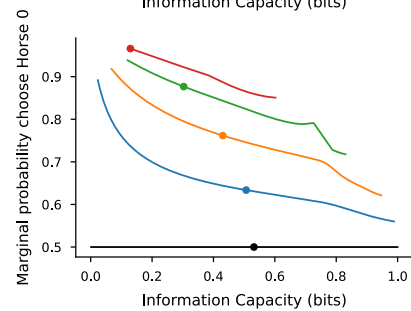
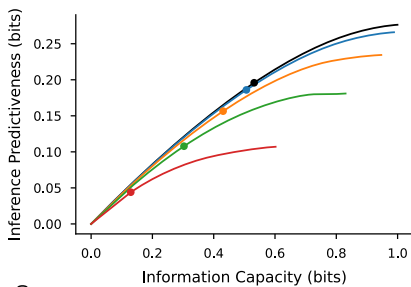
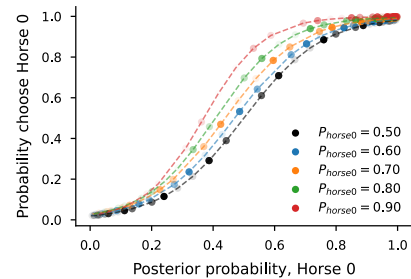
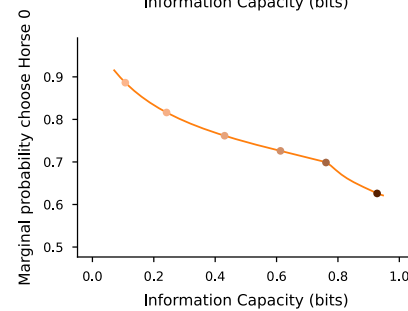
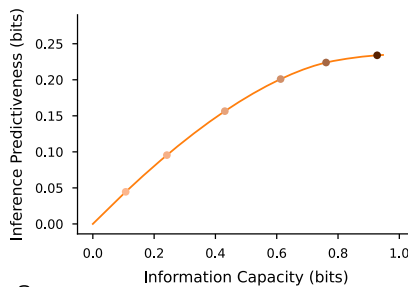
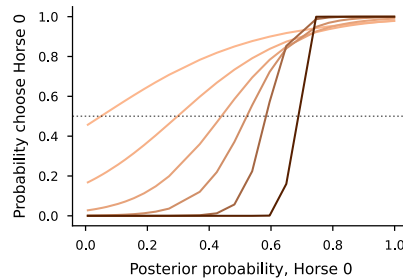
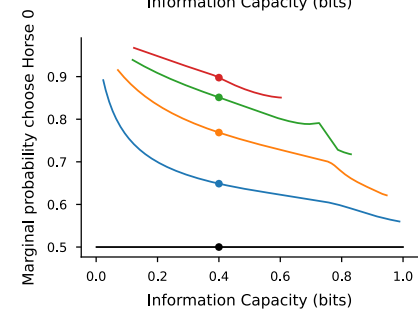
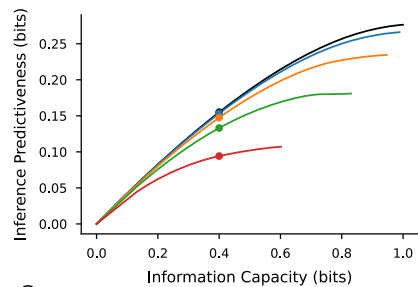
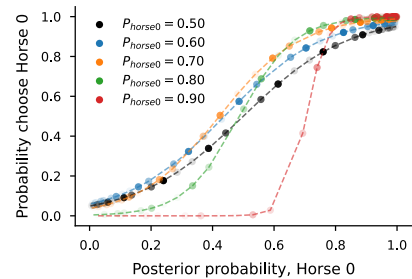
Participant with unusually high bias (red X above)





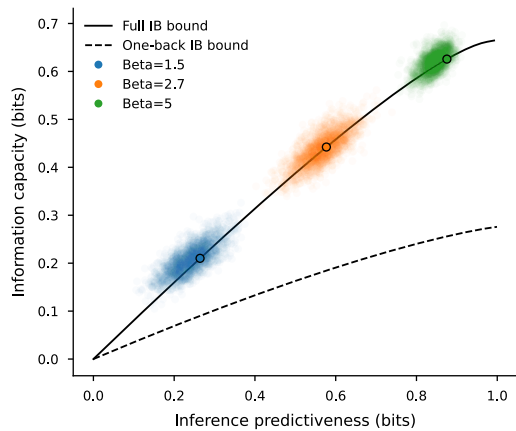
D

Jar 0 $\rightarrow$ Jar 1 hazard rate	Jar 1 $\rightarrow$ Jar 0 hazard rate	Marginal probability of Jar 0
0.01	0.01	0.50
0.01	0.015	~ 0.60
0.01	0.023	~ 0.70
0.01	0.04	~ 0.80
0.01	0.086	~ 0.90

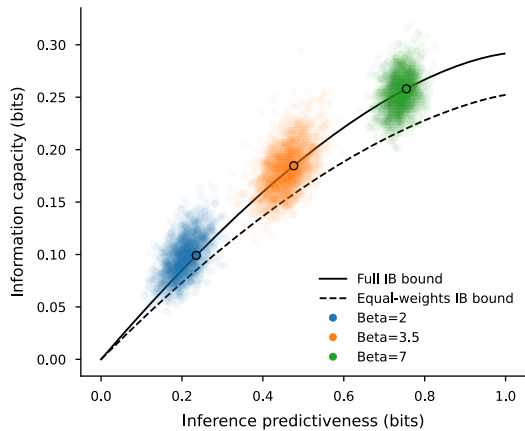
AFixed β
 $\beta = 3.5$ **B**Fixed asymmetry
 $P_{\text{prior0}} = 0.70$ **C**Fixed information capacity
 $I(X;R) = 0.4$ 

A

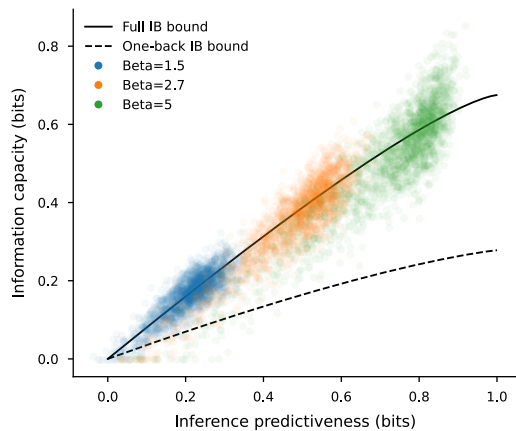
Bead prediction task
 Single trial set (500 bead draws)
 2000 choice set samples (fully optimal model)
Error from choice sampling

**C**

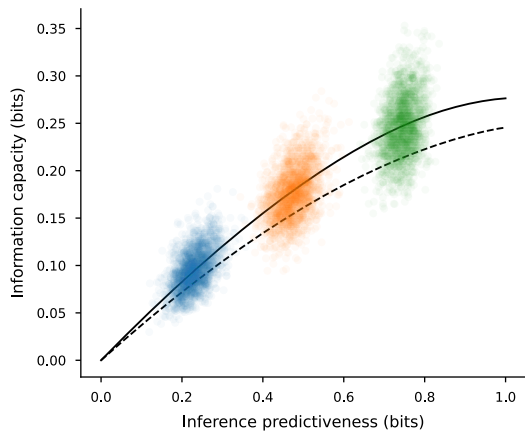
Horse prediction, intermediate W-S ratio (2.5:1)
 Single trial set (600 horse races)
 2000 choice set samples (fully optimal model)
Error from choice sampling

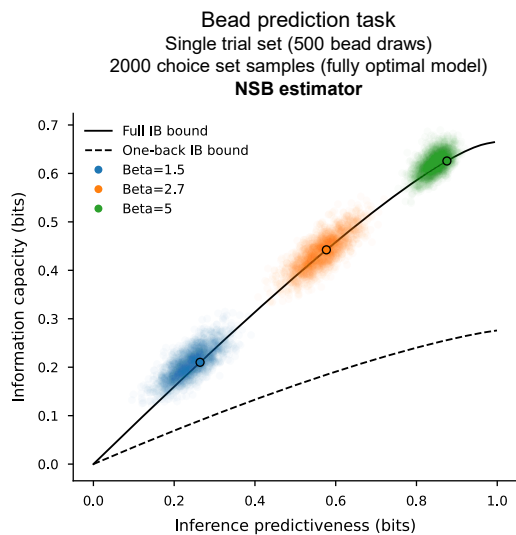
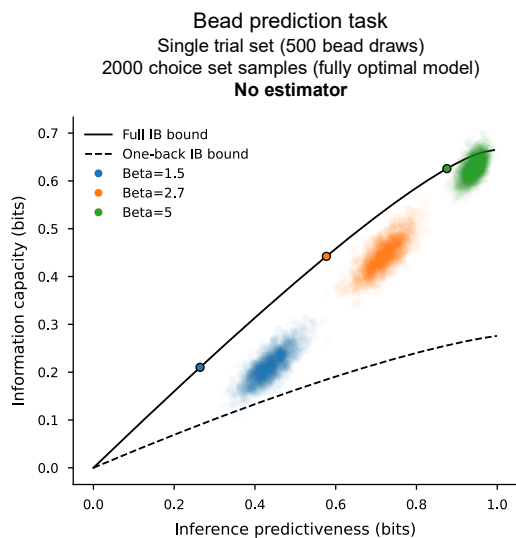
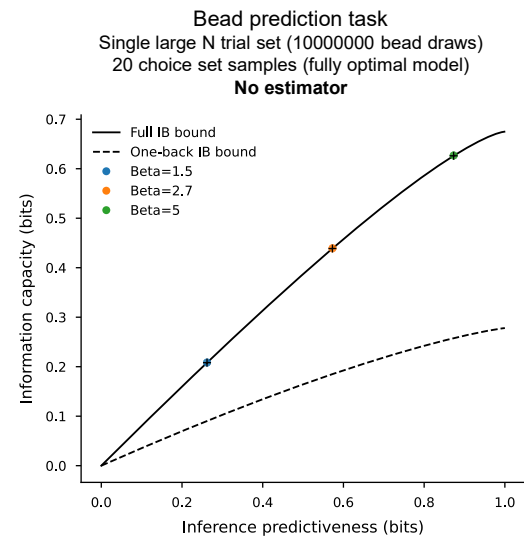
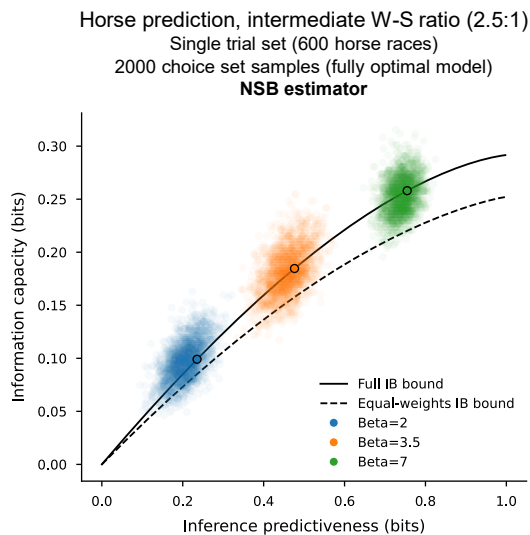
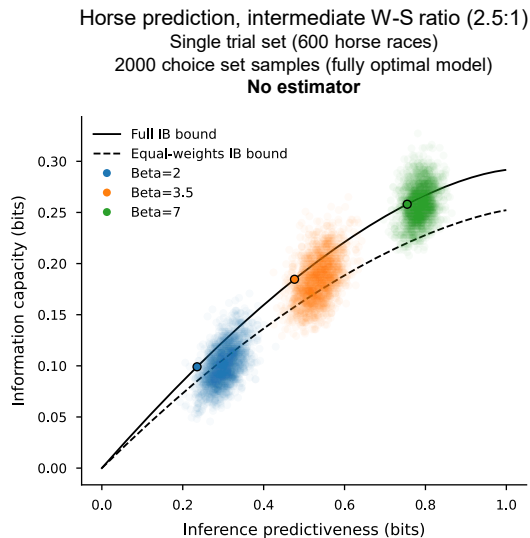
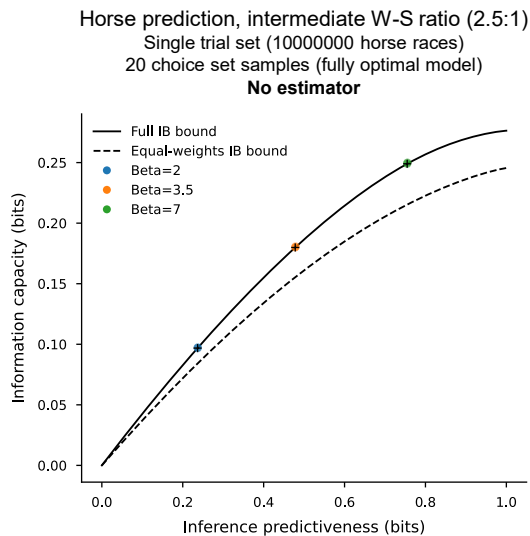
**B**

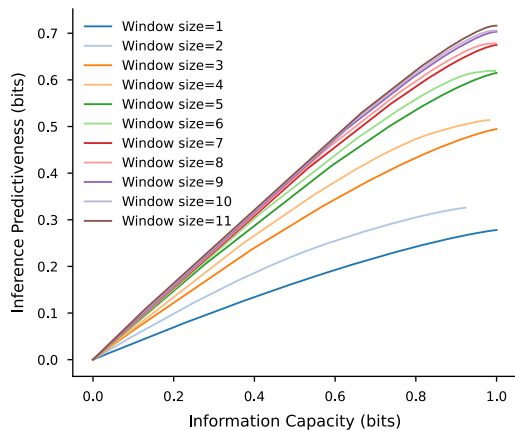
Bead prediction task
 2000 different trial sets (500 bead draws)
 1 choice set sample per trial set (fully optimal model)
Error from choice sampling + trial set sampling

**D**

Horse prediction, intermediate W-S ratio (2.5:1)
 Single trial set (600 horse races)
 2000 choice set samples (fully optimal model)
Error from choice sampling + trial set sampling



A**B****C****D****E****F**

ABead prediction task, IB bounds for theoretical $p(X,Y)$ **B**Bead prediction task, IB-optimal choice probabilities for theoretical $p(X,Y)$ applied to actual trial set (500 bead draws)